

BRIAN KIPKEMBOI

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 Cairns City, QLD

ABOUT MYSELF

Detail-oriented and analytically driven Data Science professional with professional experience spanning healthcare data systems, statistical analysis, program evaluation, and community services. Currently completing a Master of Data Science (Professional) at James Cook University with expertise in machine learning, natural language processing, computer vision, and applied analytics. Experienced in building end-to-end data pipelines, deploying ML models on edge devices, and applying NLP techniques to real-world datasets. Demonstrates strong capability in designing data collection frameworks, analysing complex datasets, and translating statistical outputs into actionable insights. Skilled in Python, R, SQL, PyTorch, and modern data science tools. Background in frontline homelessness case management and disability support brings a uniquely human-centred lens to data-driven problem solving.

CORE COMPETENCIES

- Statistical Analysis & Modelling
- Healthcare Data Systems (HMIS, DHIS, EHR)
- Data Collection & Survey Design
- Data Cleaning, Validation & Transformation
- Quantitative & Qualitative Analysis
- Reporting & Data Visualisation
- Monitoring & Evaluation (M&E)
- Research Methodology
- Database Management
- Evidence-Based Decision Support
- Machine Learning & Predictive Modelling
- Natural Language Processing & Sentiment Analysis
- Computer Vision & IoT Systems
- Web Scraping & Data Pipeline Engineering
- Deep Learning & Transformer Architectures
- Homelessness Case Management & Crisis Intervention

PROFESSIONAL EXPERIENCES

AI Research Intern

JCU Innovation & Industry Lab - Mind Scrapers Team, Cairns | February 2026 - Present

Roles:

- Completed WIL placement (EG5300) under Dr. Samantha J. Horseman, delivering 200 hours of applied AI research across five concurrent industry projects.

- Built and deployed a real-time computer vision system using YOLOv8 and OpenCV for agricultural quality detection on Raspberry Pi edge devices with LoRa/Meshtastic IoT telemetry.
- Developed a live Flask dashboard displaying quality percentages, stock levels, and sales inference from edge-transmitted sensor data.
- Conducted full NLP sentiment analysis pipeline (VADER, TextBlob, LDA topic modelling) on 1,200+ social media comments with web scraping via Apify.
- Built three progressive GPT-style transformer language models from scratch in PyTorch, including RAG retrieval and web-scraping capabilities.
- Collaborated with industry partners including Wärtsilä (maritime AI) and presented at investment workshops and the CAVE 3.0 immersive lab launch.

Homelessness Case Manager & Residential Youth Worker

Anglicare Far North Queensland, Cairns| May 2024 - Present

Roles:

- Delivered holistic, end-to-end case management to individuals and young people experiencing or at risk of homelessness across community and residential settings.
- Conducted comprehensive intake assessments, strengths-based assessments, risk assessments and safety planning in accordance with legislation and service agreements.
- Advocated on behalf of clients with housing providers, specialist homelessness services, mental health, AOD, legal and domestic and family violence services.
- Managed crisis situations and behavioural escalations using de-escalation techniques and sound professional judgement within established guidelines.
- Maintained accurate, confidential and timely documentation within approved systems in line with organisational policies, quality standards, and KPI-driven frameworks.

Disability Support Worker

Enabling Pathways, Cairns| October 2023 - June 2024

Roles:

- Collected and recorded client health and behavioural data systematically, ensuring accuracy and consistency for monitoring progress and supporting evidence-based healthcare planning decisions.
- Analysed daily client activity patterns and behavioural trends to assist healthcare professionals in evaluating care effectiveness and improving personalised care strategies.
- Maintained structured documentation systems, ensuring compliance with healthcare regulations, data integrity standards, and organisational reporting requirements for audit and review purposes.
- Collaborated with multidisciplinary healthcare teams to interpret client data, contributing to development and adjustment of individualised care and support plans.
- Monitored nutritional intake, PEG feeding records, and health indicators, identifying patterns and reporting anomalies to support clinical decision-making processes.
- Applied data recording and reporting techniques to track client outcomes, supporting evaluation of service quality and continuous improvement initiatives.
- Assisted in compiling client progress summaries and reports, translating observational and recorded data into meaningful insights for healthcare providers.
- Ensured accurate data capture during daily care activities, supporting statistical tracking of client wellbeing indicators and service delivery outcomes.

Statistician

Nuru International, Migori| November 2022 - September 2023

Roles:

- Designed and developed survey instruments, questionnaires, and structured data collection frameworks to ensure statistically valid, reliable, and representative datasets for analysis and reporting.
- Collected, processed, validated, and managed large datasets from multiple sources, ensuring high standards of data quality, accuracy, consistency, and integrity across all statistical operations.
- Performed advanced statistical analyses including regression, hypothesis testing, confidence intervals, and multivariate modelling to generate meaningful insights supporting evidence-based decision-making.
- Utilised statistical software such as SPSS and STATA to conduct complex data manipulation, predictive modelling, and quantitative analysis for program evaluation and research purposes.
- Prepared analytical reports, dashboards, charts, and data visualisations using Excel and other tools to effectively communicate statistical findings to technical and non-technical stakeholders.
- Monitored and improved data collection processes by performing data cleaning, coding, validation, and error-checking to ensure reliable datasets suitable for accurate statistical interpretation.
- Supported project planning, monitoring, and evaluation activities by providing data-driven insights and recommendations to enhance program performance and strategic outcomes.
- Maintained comprehensive documentation of statistical methodologies, procedures, and outputs while ensuring compliance with ethical standards, data confidentiality, and organisational reporting requirements.

Statistician

Reale Hospital, Eldoret| November 2021 - November 2022

Roles:

- Designed and implemented statistical studies, surveys, and observational research to support hospital decision-making, operational planning, and evaluation of healthcare programs and services.
- Collected, validated, and cleaned healthcare datasets, ensuring data accuracy, completeness, and reliability for statistical analysis and reporting within hospital information systems.
- Developed and managed health databases, organizing large volumes of patient and operational data to support efficient analysis, reporting, and evidence-based decision-making processes.
- Applied advanced statistical techniques including regression analysis, hypothesis testing, and trend analysis to interpret complex healthcare data and identify key performance indicators.
- Prepared comprehensive statistical reports, visualizations, and summaries to effectively communicate findings and insights to hospital management and multidisciplinary stakeholders.
- Provided data-driven recommendations to support policy development, operational improvements, and strategic planning aimed at enhancing patient care and service delivery.
- Utilised statistical software such as R, Python, and SPSS to perform data analysis, predictive modelling, and ensure reproducibility and accuracy of statistical results.
- Ensured compliance with data governance, confidentiality, and ethical standards while handling sensitive health information, and supported staff through training in statistical methods and data management practices.

EDUCATION & CERTIFICATIONS

Master of Data Science (Professional)| James Cook University | September 2023 - Present

Bachelor of Science in Applied Statistics| Rongo University| September 2017 - December 2021

- Second Class Honours (Lower Division)

Certificate III in Individual Support| Community Services | Completed

Certificate in Computer Software and Maintenance| Royalnet Institute, Kitale| February 2017 - June 2017

- Credit I

Kenya Certificate of Secondary Education (KCSE) | Lelmokwo High School| January 2013 - December 2016

- Score: 52 Points

Kenya Certificate of Primary Education (KCPE) | Sunrise Academy| January 2002 - December 2012

- Score: 353 / 500

TECHNICAL SKILLS

- **Statistical Tools:** R, SPSS, STATA, Python (NumPy, pandas, SciPy)
- **Data Tools:** MS Excel (Advanced), Access, Pandas, Jupyter Notebook, Google Colab, AWS SageMaker
- **Platforms & Deployment:** Git, GitHub, Netlify, Vercel, Docker, Flask, React, Streamlit
- **Systems:** DHIS2, HMIS, EHR Systems, Raspberry Pi, LoRa/Meshtastic IoT, Linux
- **Techniques:** Regression Analysis, Sampling Methods, Data Modelling, Machine Learning (Scikit-learn, XGBoost), Deep Learning (PyTorch, TensorFlow), NLP (NLTK, VADER, TextBlob, LDA), Computer Vision (OpenCV, YOLOv8), Web Scraping (BeautifulSoup, Selenium, Apify), Transfer Learning (MobileNetV2), RAG Systems, Transformer Architectures
- **Visualisation:** Tableau, Power BI, Matplotlib, Seaborn, Plotly, Streamlit Dashboards
- **Languages:** Python, R, SQL, JavaScript, HTML/CSS, Bash

PERSONAL SKILLS

- Analytical thinking and problem-solving ability
- Strong written and verbal communication skills
- Ability to interpret complex datasets accurately
- Attention to detail and data accuracy focus
- Team collaboration and stakeholder engagement
- Time management and multitasking ability
- Adaptability in dynamic healthcare environments
- Data storytelling and technical presentation
- Cross-cultural communication and stakeholder engagement

- Crisis management and de-escalation
- Rapid prototyping and end-to-end project delivery

AREAS OF INTEREST

- Biostatistics and Public Health Analytics
- Predictive Modelling and Machine Learning
- Health Data Systems and Digital Health
- Epidemiological Research
- Data-driven Policy and Decision Making
- Statistical Consulting and Research Design
- Natural Language Processing & Low-Resource Language AI
- Computer Vision for Agriculture & IoT Systems
- Civic Technology & AI for Social Impact
- Deep Learning & Transformer Model Development

REFERENCES

Available on Request